

Dr.-Ing. Ee Heng Chen

Machine Learning / AI Engineer | Agentic AI Systems, LLM Orchestration, MLOps

Munich, Germany | +49 176 32836088 | eeheng.chen@gmail.com

github.com/cheneeheng | cheneeheng.github.io | linkedin.com/in/ee-heng-chen-dr-ing

SUMMARY

- Built Large Language Model (LLM) and Generative AI (GenAI) systems that turn unstructured regulatory and technical documents into structured, reviewable outputs, with Retrieval-Augmented Generation (RAG) and evaluation pipelines.
- Rolled out n8n to production as the company-wide automation platform, now used by ~10 engineers across 40+ workflows.
- Configured MLOps toolchains (DVC, MLflow, AWS) for reproducible and auditable ML workflows.
- Personal projects across the AI agent stack (multi-agent orchestration, Model Context Protocol (MCP) tooling, replay testing) on one principle: the LLM classifies, a deterministic policy layer decides and writes the audit trail.
- Dr.-Ing. in computer science, with delivery experience across automotive, clinical and industrial settings.

EXPERIENCE

VAIVA GmbH

12/2023 - Present

Machine Learning Engineer

- Conceived and built an agentic GenAI application, working cross-functionally with the requirements engineering team, that ingests regulatory documents and drafts structured system requirements, so engineers review a draft instead of starting from a blank page. Designed the architecture, expert-authored agent skills, deterministic format checks and the Claude Code agent harness integration.
- Deployed n8n to production as the company automation platform and led adoption across ~10 engineers, with 40+ workflows in use and a shared workflow library versioned in GitLab.
- Designed and delivered an LLM-based internal testing tool using LangChain and Gradio, incorporating RAG, structured output parsing, prompt engineering and evaluation pipelines; led it from proof of concept to functional prototype.
- Integrated AWS Bedrock as the LLM backend, giving applications API access to cloud-hosted foundation models.
- Set up MLOps toolchains for model training (data versioning with DVC, experiment tracking with MLflow) and integrated them into the company cloud environment with the DevOps team.

Munich Institute of Robotics and Machine Intelligence

04/2021 - 08/2023

Postdoctoral Researcher - Machine Learning

- Coordinated researchers and clinical stakeholders at the German Heart Center Munich, and authored two successful ethics proposals for patient data collection.
- Modeled patient and clinician actions in intensive care unit (ICU) settings using graph convolutional networks (GCNs) in PyTorch.
- Built the supporting data pipeline: an ICU data-capturing system on Raspberry Pi and Intel RealSense devices, plus a Django web application for visualization.

BMW Group

09/2017 - 03/2021

Doctoral Researcher - Machine Learning / Computer Vision (ADAS)

- Developed a vision-based system determining whether a traffic junction is safe to cross for driver assistance and self-driving vehicles; trained and evaluated CNNs in TensorFlow.
- Used Docker and Kubernetes to manage training and evaluation jobs on compute clusters, and maintained the compute infrastructure for research and prototyping.
- Published 5 conference papers; supervised 2 interns and 4 student theses.

PROJECTS

pr-compliance-gate — Policy-driven risk classification pipeline

- LLM classification and risk scoring feeding a deterministic policy table, producing auditable pass/fail decisions. Python.

mcp-cassette — Record-and-replay for agent testing

- Deterministic capture and replay of Model Context Protocol interactions for reproducible agent tests. Published on PyPI.

mlops-incident-commander — Automated model monitoring and remediation

- Multi-agent system supervising a live ML model: monitoring, investigation, remediation and post-mortem generation. Python.

presidio-compliance-stack — Data-protection compliance layer

- PII detection, audit and redaction built to Malaysia's PDPA, packaged for SMEs running LLM tooling over customer data. Python, Microsoft Presidio.

EDUCATION

Technical University of Munich

09/2017 - 07/2022

Dr.-Ing. (Ph.D.) Computer Science: Department of Informatics

- Thesis: Development of Vision-based Driving Strategies for Traffic Junction Scenarios.

Technical University of Munich

04/2015 - 08/2017

M.Sc. Robotics, Cognition, and Intelligence: Department of Informatics

Heilbronn University of Applied Sciences

04/2011 - 03/2015

B.Eng. Mechatronics and Microsystems: Faculty of Mechanics and Electronics

SKILLS

Programming and Tools: Python (8+ yrs), C++ (3+ yrs), Django, Gradio, Docker, Kubernetes, Git / GitLab, Linux, n8n

Artificial Intelligence & Machine Learning: Generative AI (GenAI), Large Language Models (LLMs), AI agents, multi-agent orchestration, Retrieval-Augmented Generation (RAG), prompt engineering, LLM evaluation, LangChain, AWS Bedrock, Claude Code / agent harnesses, Model Context Protocol (MCP), agent replay testing, Presidio, PyTorch, TensorFlow, MLOps (DVC, MLflow)

Computer Vision: OpenCV, object detection, semantic segmentation, optical flow, pose estimation

Languages: English (fluent), German (C1), Mandarin, Malay

Publications: 9 peer-reviewed conference papers (ITSC, ICRA, IROS), listed on [Google Scholar](#)